

A Data Aware Multi-bit Quantization Scheme for RF Signal Sensing

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Abstract—Traditional codebook-based signal quantization methods, such as k-means clustering, involve iterative sample-to-cluster assignments, leading to a computational cost that scales linearly with the number of samples which is expensive for large sample size. To address this challenge, we revisit the codebook-based quantization paradigm and propose an alternative approach where the codebook design problem is approximated by matching sample histograms. The computational cost thus depends only on the number of histogram bins, significantly reducing the overall burden. Based on this insight, we introduce a specific ℓ_0 -norm constrained formulation that aligns codebook values directly to the histogram bins, which ensure efficient representation of samples. To efficiently solve the resulting problem, we develop an alternating direction method of multipliers (ADMM)-based algorithm incorporating an ℓ_0 -norm projection operator. This scheme enables the construction of a flexible codebook that adapts to the sample histogram without costly sample-to-cluster computations. Experimental results on direct position determination and modulation classification tasks demonstrate that the proposed scheme achieves competitive performance while offering improved computational efficiency.

Index Terms—Signal quantization, ℓ_0 -norm constraint, alternating direction method of multipliers

I. INTRODUCTION

The rising demand for data-intensive radio frequency (RF) sensing applications, such as modulation sensing [1], [2], direct localization [3], and others, has led to a substantial increase in the volume of signal samples. This growth presents significant challenges for efficient RF sensing tasks particularly in bandwidth-constrained communication links, as illustrated in Fig. 1. In these scenarios, the quantization of in-phase and quadrature (IQ) signal samples emerges as a critical factor for ensuring the effectiveness of downstream sensing tasks. Codebook-based quantization (CQ) [4] provides an effective approach tailored to the statistical properties of IQ signal samples. Rather than partitioning the signal sample value range into uniform intervals, CQ quantizes each IQ signal sample by mapping it to the nearest codeword in a learned or predefined codebook. This introduces a non-uniform partitioning allowing finer representation in regions with higher probability. Consequently, CQ achieves improved rate-distortion performance over uniform quantization, particularly at low bit precision.

Recent works have started to explore adaptive or hybrid codebook-based quantization schemes tailored to specific sensing applications, such as MIMO beamforming [5], localization [6], and spectrum sensing [7]. Despite recent advances, existing methods still rely on

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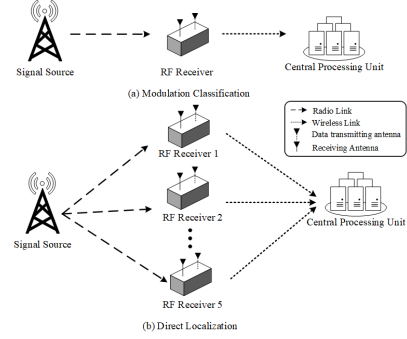


Fig. 1. Illustration of two signal sensing scenarios. Quantization is performed at the RF receiver.

scenario-specific truncation, which limits their general applicability across different sensing conditions. The distribution of signal samples is influenced by various environmental factors, like multipath propagation, hardware impairments, and etc, making it highly scenario-dependent. The development of quantization schemes that can effectively adapt to distributions has received limited attention, representing a gap in the current literature.

In this work, we propose a multi-bit quantization scheme that generates a flexible codebook tailored to the distribution of signal samples. Unlike conventional codebook searching algorithms, such as k-means clustering, which rely on computationally intensive optimization of sample-to-cluster associations and are sensitive to outliers, our scheme reformulates the problem as an optimization task with a specific ℓ_0 -norm constraint. This formulation adapts the codebook to the sample histogram, with the optimization variables limited to the codebook values assigned to histogram bins. Leveraging a projection operator for the ℓ_0 -norm, an efficient alternating direction method of multipliers (ADMM)-based algorithm is subsequently developed.

II. PROBLEM FORMULATION

A. Preliminaries

Denote vector $\mathbf{s} = (s_1, s_2, \dots, s_N) \in \mathbb{R}^N$ as the collected signal samples of size N and $\mathcal{C} = \{c_1, c_2, \dots, c_B\} \subseteq \mathbb{R}^B$ be a learnable codebook, the CQ scheme for wireless signal can be in general formulated as

$$\min_{\mathcal{C}, \{z_n\}} \sum_{n=1}^N |s_n - c_{z_n}|^2 \quad (1)$$

where $z_n \in \{1, \dots, B\}$ denotes the index of the codebook entry assigned to signal sample s_n [8]. For T -bit budget, the choice of B usually satisfies $\log_2 B \leq T$ since a part of codewords are often reserved for verification or other purposes in practice [9].

A typical algorithm to tackle Problem (1) is the k-means [10] algorithm, which is known to be sensitive to outliers. This sensitivity

arises because k-means fits the original signal, and the outliers dominate the objective function. As a result, centroids are often positioned away from the majority of samples, leading to inefficient use of the codebook budget. A toy example is presented in Fig. 2, where 1% of the signal samples are randomly selected using a uniform distribution and perturbed with outliers. Under this scenario, k-means produces uncontrolled cluster centers and poorly defined cluster boundaries. In addition to the outlier sensitivity, the computational cost of k-means per iteration scales linearly with the number of samples, i.e., $O(NB)$, which make it not applicable in practice for N being huge. An alternative way is to directly learn the cluster boundaries

$$\mathcal{T} = \{\tau_0, \tau_1, \dots, \tau_{B-1}, \tau_B\},$$

where $\tau_0 < \tau_1 < \dots < \tau_B$. This enables to establish an explicit relationship between sample values and the codebook, while simultaneously providing greater flexibility in the selection of cluster centers. For simplicity, we enforce the interval centers as the codeword, i.e., $c_b = \frac{1}{2}(\tau_{b-1} + \tau_b)$, $b \in \{1, \dots, B\}$, then Problem (1) can be re-casted as

$$\begin{aligned} \min_{\mathcal{T}} \quad & \sum_{n=1}^N |s_n - c_{z_n}|^2 \\ \text{s.t.} \quad & z_n = b \text{ if } s_n \in I_b \triangleq [\tau_{b-1}, \tau_b), \forall n. \end{aligned} \quad (2)$$

Comparing Problems (1) and (2), the critical distinction lies in the treatment of the assignment $\{z_n\}$. In Problem (2), these variables are implicitly eliminated by directly defining the cluster boundaries. Consequently, there are only $B + 1$ optimization variables, i.e., τ_b , $b = 0, 1, \dots, B$, which allows us to develop efficient algorithm later, especially for N being huge.

B. Proposed Optimization Formulation

Notice that the objective in (2) can be further re-expressed as

$$\sum_{b=1}^B \sum_{s_n \in I_b} |s_n - c_b|^2,$$

which aims at aligning each signal sample s_n within interval I_b using a unique code c_b . Finding $\{I_b, c_b\}$ still requires visiting all samples which may be costly. For distinct samples s_m and s_n , if s_m is close to s_n enough, then the same c_b is applied to encode them. This motivates us to approximate the objective using the histogram of s . Note that the size of the histogram is controllable, which is usually much less than the number of samples. This brings convenience for later developing efficient algorithm for optimizing codebook $\mathcal{C}$.

Formally, denote $\mathbf{h} = (h_1, h_2, \dots, h_K) \in \mathbb{R}^K$ ($K > B$) as the histogram representation of s (c.f., example in Fig. 3) and let ν_k be the sample value associated with h_k , where signal samples within $\mathcal{A}_k = [\nu_{k-1}, \nu_k)$ are counted in h_k ¹. Accordingly, the cluster boundary τ_b in (2) can be mapped to histogram bin indices via

$$\kappa_b - 1 = \arg \min_k |\nu_k - \tau_b|. \quad (3)$$

Thus we can define interval for histogram indices as $\mathcal{I}_b = [\kappa_{b-1}, \kappa_b)$. Then, by introducing a vector $\mathbf{x} = (x_1, x_2, \dots, x_K) \in \mathbb{R}^K$, Problem (2) is approximated by using histogram $\mathbf{h}$ as

$$\begin{aligned} \min_{\{\kappa_b\}, \mathbf{x}} \quad & \frac{1}{2} \sum_{k=1}^K |x_k - h_k|^2 \\ \text{s.t.} \quad & x_k = x_{k'}, \quad k, k' \in \mathcal{I}_b. \end{aligned} \quad (4)$$

¹Denote $s_{\min}$ and $s_{\max}$ as the predefined minimal and maximal sample values, then region $[s_{\min}, s_{\max}]$ is uniformly partitioned into K bins, each $\mathcal{A}_k = [\nu_{k-1}, \nu_k)$, $\nu_k = s_{\min} + \frac{k}{K}(s_{\max} - s_{\min})$. The normalized histogram $h_k = \frac{1}{N} \sum_{n=1}^N \mathbf{1}\{s_n \in \mathcal{A}_k\}$, where $\mathbf{1}\{s_n \in \mathcal{A}_k\}$ is the indicator function of $\mathcal{A}_k$.

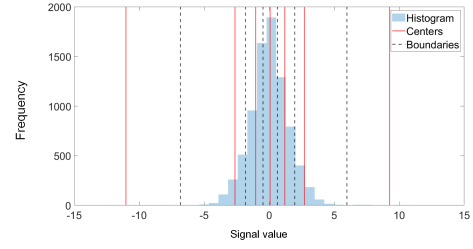


Fig. 2. Quantization results of k-mean in the presence of outliers.

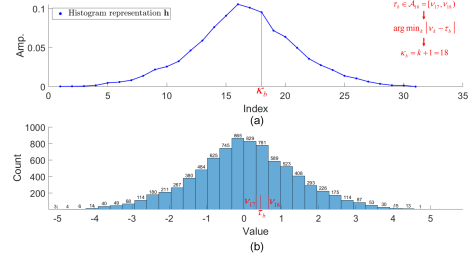


Fig. 3. (a) The histogram representation $\mathbf{h}$. (b) The original histogram.

Notice that κ_b in (4) actually indicates the changing indices of x_k , i.e., $x_{\kappa_b} \neq x_{\kappa_b-1}$ and the presence of $\{\kappa_b\}$ can be further eliminated by utilizing ℓ_0 norm (see Proposition 1). Moreover, robustness to outliers can be enforced when constructing the histogram, where outliers can be counted within $\mathcal{A}_1$ or $\mathcal{A}_K$.

By Proposition 1, Problem (4) can be equivalently transformed into a compact form as

$$\begin{aligned} \min_{\mathbf{x}} \quad & \frac{1}{2} \|\mathbf{x} - \mathbf{h}\|^2 \\ \text{s.t.} \quad & \|\mathbf{M}\mathbf{x}\|_0 \leq B - 1, \end{aligned} \quad (5)$$

where the dimension of optimization variable $\mathbf{x}$ is significantly smaller than the sample size N . This dimensionality reduction leads to a major computational advantage, as the per-iteration complexity of solving Problem (5) depends only on K , whereas traditional k-means clustering has a computational complexity that scales with N . Furthermore, the detailed procedure for constructing the codebook c_b from the optimized variable $\mathbf{x}$ is provided in Section III-B.

Proposition 1. Given $1 = \kappa_0 < \kappa_1 < \dots < \kappa_{B-1} < \kappa_B = K + 1$, denote set $\mathcal{X} := \{\mathbf{x} \in \mathbb{R}^K : x_k = x_{k'}, \forall k, k' \in \mathcal{I}_b\}$. Then, $\mathcal{X}$ can also be equivalently expressed as

$$\mathcal{X} = \{\mathbf{x} \in \mathbb{R}^K : \|\mathbf{M}\mathbf{x}\|_0 \leq B - 1\},$$

where $\mathbf{M} \in \mathbb{R}^{(K-1) \times K}$ represents a row-full-rank matrix:

$$\mathbf{M}_{i,j} = \begin{cases} 1, & \text{if } i = j, \\ -1, & \text{if } i = j - 1, \quad (1 \leq i \leq K - 1, 1 \leq j \leq K) \\ 0, & \text{otherwise.} \end{cases}$$

Proof. Define sets $\mathcal{X}_1 \triangleq \{\mathbf{x} \in \mathbb{R}^K : x_k = x_{k'}, \forall k, k' \in \mathcal{I}_b\}$ and $\mathcal{X}_2 \triangleq \{\mathbf{x} \in \mathbb{R}^K : \|\mathbf{M}\mathbf{x}\|_0 \leq B - 1\}$, where $\mathbf{M}\mathbf{x} = (x_2 - x_1, x_3 - x_2, \dots, x_K - x_{K-1})^\top$. Next, we show $\mathcal{X}_1 = \mathcal{X}_2$. Suppose $\mathbf{x} \in \mathcal{X}_1$, i.e., for $b = 1, \dots, B$, $x_k = x_{k'}, \forall k, k' \in \mathcal{I}_b$. This implies that for any $k \notin \{\kappa_1 - 1, \kappa_2 - 1, \dots, \kappa_{B-1} - 1\}$, $x_{k+1} - x_k = 0$. The nonzero entries in $\mathbf{M}\mathbf{x}$ are at indices $k = \kappa_b - 1$ for $b = 1, \dots, B - 1$. Thus, $\mathbf{M}\mathbf{x}$ has at most $B - 1$ nonzero entries, i.e., $\|\mathbf{M}\mathbf{x}\|_0 \leq B - 1$. Hence, $\mathbf{x} \in \mathcal{X}_2$ and $\mathcal{X}_1 \subseteq \mathcal{X}_2$. Conversely, suppose $\mathbf{x} \in \mathcal{X}_2$, i.e.,

$\|\mathbf{M}\mathbf{x}\|_0 \leq B - 1$. Let $d_1 < d_2 < \dots < d_J$ ($J \leq B - 1$) be the indices where $(\mathbf{M}\mathbf{x})_{d_j} \neq 0$. Define $\kappa_0 = 1$, $\kappa_1 = d_1 + 1$, $\kappa_2 = d_2 + 1$, ..., $\kappa_J = d_J + 1$, and $\kappa_{J+1} = K + 1$, then for any k in $[\kappa_{b-1}, \kappa_b)$, we have $x_{k+1} - x_k = 0$. Hence $\mathbf{x} \in \mathcal{X}_1$ and $\mathcal{X}_2 \subseteq \mathcal{X}_1$. Since $\mathcal{X}_1 \subseteq \mathcal{X}_2$ and $\mathcal{X}_2 \subseteq \mathcal{X}_1$, we have $\mathcal{X}_1 = \mathcal{X}_2$. This completes the proof. $\square$

III. PROPOSED ADMM-BASED ALGORITHM

The difficulty of solving Problem (5) lies in the nonconvex and non-smooth ℓ_0 norm constraint over $\mathbf{M}\mathbf{x}$. Realizing that projection onto ℓ_0 norm ball is just selecting the few largest entries, Problem (5) thus can be effectively tackled by variable splitting trick. In particular, by introducing auxiliary variable $\mathbf{z} = \mathbf{M}\mathbf{x} \in \mathbb{R}^{K-1}$, Problem (5) can be reformulated as

$$\begin{aligned} \min_{\mathbf{x}} \quad & \frac{1}{2} \|\mathbf{x} - \mathbf{h}\|^2 \\ \text{s.t.} \quad & \mathbf{M}\mathbf{x} = \mathbf{z}, \quad \|\mathbf{z}\|_0 \leq B - 1. \end{aligned} \quad (6)$$

In Problem (6), the ℓ_0 norm constraint is applied solely to $\mathbf{z}$, enabling the development of ADMM-based algorithm to address it.

A. ADMM for Solving Problem (6)

Denote L_ρ as the augmented Lagrangian function for Problem (6)

$$L_\rho(\mathbf{x}, \mathbf{z}, \lambda) = \frac{1}{2} \|\mathbf{x} - \mathbf{h}\|^2 + \lambda^\top (\mathbf{M}\mathbf{x} - \mathbf{z}) + (\rho/2) \|\mathbf{M}\mathbf{x} - \mathbf{z}\|^2$$

where $\lambda \in \mathbb{R}^{K-1}$ and $\rho > 0$ is the penalty parameter. At the k -th iteration, the primal and dual steps of ADMM are derived as follows.

1) *Updating $\mathbf{x}$* : Fix $\mathbf{z}^k$ and λ^k , $\mathbf{x}^{k+1}$ is updated as

$$\begin{aligned} \mathbf{x}^{k+1} &= \underset{\mathbf{x}}{\operatorname{argmin}} L_\rho(\mathbf{x}, \mathbf{z}^k, \lambda^k) \\ &= \underset{\mathbf{x}}{\operatorname{argmin}} \frac{1}{2} \|\mathbf{x} - \mathbf{h}\|^2 + \lambda^{k\top} (\mathbf{M}\mathbf{x} - \mathbf{z}^k) + \frac{\rho}{2} \|\mathbf{M}\mathbf{x} - \mathbf{z}^k\|^2 \end{aligned}$$

Since this subproblem is strongly convex with quadratic objective, $\mathbf{x}^{k+1}$ has a closed-form given as

$$\mathbf{x}^{k+1} = (\mathbf{E} + \rho \mathbf{M}^\top \mathbf{M})^{-1} (\mathbf{h} - \mathbf{M}^\top \lambda^k + \rho \mathbf{M}^\top \mathbf{z}^k) \quad (7)$$

2) *Updating $\mathbf{z}$* : Fix $\mathbf{x}^{k+1}$ and λ^k , $\mathbf{z}^{k+1}$ is updated as

$$\mathbf{z}^{k+1} = \underset{\mathbf{z}}{\operatorname{argmin}} L_\rho(\mathbf{x}^{k+1}, \mathbf{z}, \lambda^k) \quad \text{s.t.} \quad \|\mathbf{z}\|_0 \leq B - 1$$

By a straightforward derivation, this subproblem corresponds to an ℓ_0 norm projection problem given as

$$\begin{aligned} \mathbf{z}^{k+1} &= \underset{\|\mathbf{z}\|_0 \leq B-1}{\operatorname{argmin}} \left\| \mathbf{z} - \frac{1}{\rho} (\lambda^k + \rho \mathbf{M}\mathbf{x}^{k+1}) \right\|^2 \\ &= \operatorname{Proj}_{B-1} \left(\frac{1}{\rho} (\lambda^k + \rho \mathbf{M}\mathbf{x}^{k+1}) \right) \end{aligned} \quad (8)$$

where $\operatorname{Proj}_r(\cdot)$ denotes the projection onto the ℓ_0 norm ball with radius r , i.e., retaining the r largest absolute values while setting all remaining elements to zero.

3) *Updating λ* : Fix $\mathbf{x}^{k+1}$ and $\mathbf{z}^{k+1}$, λ^{k+1} is updated as

$$\lambda^{k+1} = \lambda^k + \rho (\mathbf{M}\mathbf{x}^{k+1} - \mathbf{z}^{k+1}). \quad (9)$$

The complete procedure for solving problem (6) is described in Algorithm 1. The dominated computational complexity per iteration is the matrix multiplication of order $O(K^2)$, and by exploiting the sparse structure of matrix $\mathbf{M}$, it can be reduced to $O(K \log K)$ using Thomas algorithm [11]. In comparison, the complexity of k-means per iteration is $O(NB)$, which is much larger under practical setting, e.g., for one second signal samples with 1 MHz sampling rate, $N = 10^6$ and $K \sim 10$. Finally, it is worth noting that convergence is

Algorithm 1 ADMM-based algorithm for solving problem (6)

- 1: **Initialize** $\mathbf{x}^1 = \mathbf{0}_K$, $\mathbf{z}^1 = \mathbf{0}_{(K-1)}$, $\lambda^1 = \mathbf{1}_{(K-1)}$, $\rho = 10^2$
- 2: **repeat**
- 3: Update $\mathbf{x}^{k+1}$ via (7);
- 4: Update $\mathbf{z}^{k+1}$ via (8);
- 5: Update λ^{k+1} via (9);
- 6: $k \leftarrow k + 1$;
- 7: **until** $\frac{\|\mathbf{x}^{k+1} - \mathbf{h}\|^2 - \|\mathbf{x}^k - \mathbf{h}\|^2}{\|\mathbf{x}^k - \mathbf{h}\|^2} \leq \epsilon$.
- 8: **Output:** $\mathbf{x}^{k+1}$

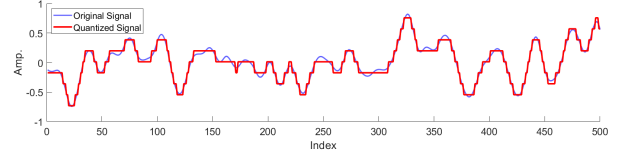


Fig. 4. The quantization results of real-world recorded signal in direct position determination (DPD) task with $B=16$.

guaranteed for sufficiently large values of ρ . For further details, we refer readers to [12] and details are not repeated here.

B. Quantize Signal using $\mathbf{x}$

Next, we present a procedure for constructing quantized signal from the optimized $\mathbf{x}$. Note that the indices $\{\kappa_b\}_{b=0}^B$ can be directly extract from $\mathbf{x}$, which indicates changing boundaries. Thus we can construct value intervals as $\mathcal{B}_b = [\tau_{b-1}, \tau_b)$, where $\tau_b = \nu_{\kappa_b-1}$. Define codebook $c_b = \frac{\tau_{b-1} + \tau_b}{2}$, then s_n is quantized as

$$\hat{s}_n = \begin{cases} c_b, & \text{if } s_n \in \mathcal{B}_b \text{ for some } b, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

Note that restricting c_1 or c_B can also introduce robustness to outliers. A toy example of the quantized signal is illustrated in Fig. 4, the quantized signal closely approximates the shape of the original signal, which demonstrates the validation of this recovery procedure.

IV. EXPERIMENTAL RESULTS

A. Experimental Setup

We evaluate the performance of the proposed quantization scheme on two downstream sensing tasks, one is the DPD task [3], [13] and the other is modulation classification (MC) task [1]. Four quantization methods are selected as baselines, namely uniform quantization [14], vector quantization [15], [16], k-means clustering [10], and equal-mass quantile quantization [17]. All experiments are conducted on a workstation with an Intel Core i7-13650HX CPU. It is worth noting that quantization parameters are strictly application-dependent, typically dictated by hardware constraints and available bandwidth. Our proposed method is designed to support arbitrary bit-depth quantization, ensuring flexibility across diverse sensing scenarios.

TABLE I. Positions of RF receivers (Rx) and transmitter (Tx)

Node	x (km)	y (km)	Node	x (km)	y (km)	Node	x (km)	y (km)
Rx 1	9.922	113.253	Rx 3	10.063	112.404	Rx 5	10.486	112.845
Rx 2	10.066	113.017	Rx 4	10.705	113.563	Tx	11.382	113.195

For the DPD task, real-world signal samples are collected using five distributed single antenna RF receiver (c.f., Fig. 4). There is one emitter transmits a QPSK modulation signal with bandwidth 200 kHz at 1.42 GHz carrier frequency. The locations of receivers and transmitter are given in Table I. At each receiver (synchronized by GPS), signal is sampled at 1.28 MHz over a duration of 93.75 ms,

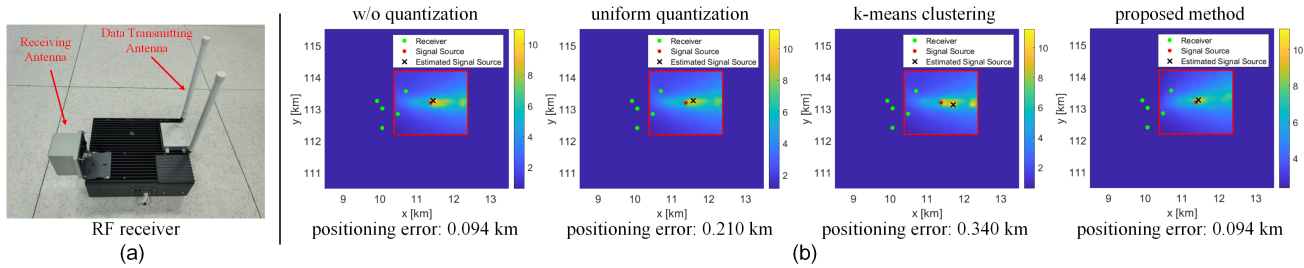


Fig. 5. (a) Photograph of RF receiver (Rx). (b) Positioning error obtained using different quantization methods ($B = 6$).

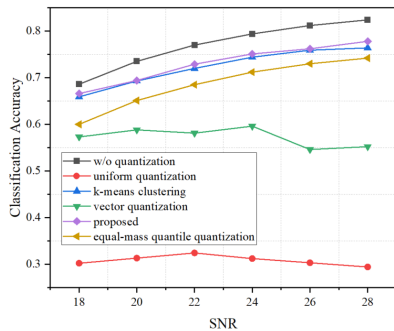


Fig. 6. Accuracy across different SNR conditions ($B = 32$).

resulting $120,000 \times 5$ signal samples in total. A $2 \text{ km} \times 2 \text{ km}$ search area (divided into a 15×15 grids) is employed for localization. The signal samples from five receivers are quantized individually and sent to a central processing unit for position estimation.

For the MC task, we generate 200 frames of signal samples at carrier frequency of 902 MHz with a sampling rate of 200 kHz. Each frame contains 1024 symbols, modulated by one of three formats: QPSK, 8PSK, and PAM4. After oversampling by a factor of 8 (zero-stuffing), this yields 8192 complex samples per frame (≈ 40.96 ms duration). The symbols are pulse-shaped by a root-raised cosine filter and transmitted through a Rician fading channel with default setting [18]. Additive white Gaussian noise is added to produce SNR values $\{18, 20, \dots, 28\}$ dB. The 200 frames are classified individually by the pretrained CNN classifier [19], and the overall classification accuracy is computed as the average across all frames.

B. Quantitative Results

The results of the DPD task with $B = 6$ (3-bit quantization) are presented in Fig. 5 and Table II, where two codewords are specifically reserved for error indication. For position estimation, we select the grid point with the highest value as the estimated location. Notably, our proposed method achieves localization accuracy equivalent to that of using unquantized signal samples, whereas all baseline methods exhibit significantly larger errors. The poor performance of k-means is attributed to the presence of outliers in the recorded signal samples, which negatively impact its clustering accuracy. Furthermore, our method demonstrates a significantly lower runtime compared to k-means, owing to its reduced dimensionality and computational efficiency, even when the entire pipeline (including the histogram operation) is accounted for.

The results of the MC task with 5-bit quantization are shown in Fig. 6 and Table III. Across all SNR settings, our proposed scheme delivers competitive performance relative to k-means and consistently outperforms other baseline methods. In terms of computational cost,

our approach is notably more efficient, requiring substantially less runtime than the other baselines.

TABLE II. Comparison of running time and positioning error.

Method	Overall Runtime (s)	Var. Dimension	Error (km)
Uniform quantization	0.05	—	0.210
K-means clustering	28.79	$[120000 \times 5]$	0.340
Proposed method	0.82	$[24 \times 5]$	0.094

TABLE III. Comparison of running time and accuracy.

Method	Overall Runtime (s)	Var. Dimension	Accuracy
Uniform quantization	0.04	—	0.308
K-means clustering	34.49	$[8192 \times 200]$	0.723
Vector quantization	14.07	$[8192 \times 200]$	0.573
Proposed method	2.10	$[20 \times 200]$	0.730

Next, considering the varying number of signal samples across codebooks, Huffman coding [20] is employed. It applies variable-length coding to the codebook to further improve coding efficiency. We quantize the PAM4 signal samples using both our method and k-means clustering, and then apply Huffman coding to the resulting codebooks. The total bits and classification accuracy are shown in Table IV, we see that our method saves 19.69% more bits and improves detection accuracy by 2.2% compared to k-means. This is because our scheme learns the signal distribution that can assign the same bits to signal samples that are close in value and occur with similar frequency. This aligns with the idea behind Huffman coding, which assigns shorter codewords to more frequent values and thus saves budget.

TABLE IV. Comparison of transmitted bits and accuracy.

Method / Signal	Bits	Accuracy
Original IQ signal (8192×16 -bit)	131072	0.994
3-bit quantization w/o Huffman	24576	—
3-bit quantization (K-means) w/ Huffman	21892	0.964
3-bit quantization (Proposed) w/ Huffman	17052	0.988

V. CONCLUSION

In this paper, we reformulate signal quantization as a histogram fitting problem, enabling the generation of a flexible codebook that is specifically tailored to the underlying distribution of signal samples. To solve this formulation, we develop an efficient ADMM-based algorithm that leverages an ℓ_0 -norm projection operator for adaptive codebook construction. Experimental results on both the DPD and MC tasks validate the effectiveness of our approach, demonstrating its superior computational efficiency and robust performance in comparison to traditional methods.

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